

Revenue Transactions Dataset

1. Executive Summary

This project focuses on analysing a Revenue Transaction dataset using Python to extract meaningful business insights from retail sales data. The analysis includes data cleaning, preprocessing, exploratory data analysis (EDA), statistical analysis, and data visualization to understand sales performance, customer purchasing behaviour, payment method preferences, and revenue trends. Various Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to transform raw data into actionable business insights. The findings from this project help support data-driven decision-making and improve overall business performance.

2. Project Objective

The primary objective of this project is to analyse revenue transaction data and identify key business patterns that can support strategic decision-making. The project aims to:

- Analyse sales and revenue performance using Python.
- Clean and preprocess the dataset to improve data quality.
- Perform exploratory data analysis to discover trends and patterns.
- Visualize customer purchasing behaviour and payment preferences.
- Generate business insights and recommendations based on data analysis.
- Demonstrate an end-to-end data analytics workflow using Python for retail business intelligence.

3. Data Sources

Source: Mendeley Data – Revenue Transactional Dataset

Location: Surabaya, East Java, Indonesia (Data collected from an anonymized Food & Beverage (F&B) MSME operating multiple retail outlets.)

Year/Timeline: January 2025 – September 2025

Domain: Retail Sales & Revenue Analytics (Food & Beverage Industry)

Dataset Size: 53,820 transaction records, 11 transaction features, and 96 product records across 3 retail outlets.

4. Dataset Architecture & Raw Schemas

4.1 Dataset Architecture

The Revenue Transaction dataset captures day-to-day sales transactions from multiple retail outlets. Each record represents a single customer transaction and includes information related to the transaction date and time, sales amount, tax, purchased items, payment method, and customer loyalty status. During preprocessing, additional analytical features were created to support business intelligence and trend analysis.

Stage	Description
Data Source	Revenue Transaction CSV Dataset
Data Loading	Imported using Pandas (pd.read_csv())
Raw Dataset	11 original attributes
Data Cleaning	Missing value check, duplicate removal, data type validation
Data Transformation	Date and time formatting, feature extraction

Stage	Description
Feature Engineering	Month, Year, Sales Category, Average Item Price, Customer Type
Statistical Analysis	Descriptive statistics, variance, standard deviation
Visualization	Univariate, Bivariate, and Multivariate Analysis
Business Insights	Revenue trends, customer behaviour, payment analysis

4.2 Raw Dataset Schema

Original Dataset Columns

S. No	Column Name	Data Type	Description
1	Outlet	Object (String)	Name or location of the retail outlet.
2	Date	Date	Date on which the transaction occurred.

S. No	Column Name	Data Type	Description
3	Time	Time	Time when the purchase was completed.
4	Net Sales	Float	Total sales amount before tax.
5	Tax	Float	Tax applied to the transaction.
6	Total Amount	Float	Final bill amount including tax.
7	Receipt Number	Integer/String	Unique identifier for each transaction.
8	Items	Object (String)	Name or description of purchased item(s).
9	Total Item	Integer	Total number of items purchased.
10	Payment Method	Object (String)	Mode of payment (Cash, Card, UPI, etc.).
11	Use Loyalty Card	Boolean/String	Indicates whether a loyalty card was used.

4.3 Engineered Features

The following columns were created during feature engineering to improve analysis and reporting.

Feature	Data Type	Purpose
Month	Integer	Extracted from the transaction date for monthly trend analysis.
Year	Integer	Used to analyse yearly sales performance.
Sales Category	Categorical	Classifies transactions into Low, Medium, or High sales categories.
Average Item Price	Float	Average price per item (Total Amount ÷ Total Item).
Customer type	Categorical	Categorizes customers based on loyalty card usage (Loyal / Regular).

Dataset Summary

Attribute	Value
Dataset Type	Structured Transaction Dataset
Original Features	11
Engineered Features	5
Total Features After Processing	16
Analysis Tool	Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly)
Project Domain	Retail Revenue Transaction Analysis

5. Tools & Technologies

Python (Google Colab)

- Data Import & Dataset Loading (Pandas read_csv())
- Exploratory Data Analysis (EDA) using Pandas
- Data Cleaning & Duplicate Value Removal
- Missing Value Detection & Data Quality Validation
- Date & Time Formatting and Data Type Standardization
- Feature Engineering (Month, Year, Sales Category, Average Item Price & Customer Type)
- Statistical Analysis (Mean, Median, Mode, Variance & Standard Deviation)
- Categorical Data Analysis using groupby() and value_counts()
- Revenue Trend & Sales Performance Analysis
- Univariate, Bivariate & Multivariate Analysis
- Data Visualization using Matplotlib & Seaborn
- Interactive Visualizations using Plotly Express & Plotly Graph Objects
- Business Insight Generation & Data-Driven Recommendations

6. Data Pre-Processing (Python / Pandas)

Data preprocessing is a crucial step in the data analysis pipeline, ensuring that the dataset is accurate, consistent, and ready for meaningful analysis. Using Python and the Pandas library, the raw revenue transaction data was cleaned, validated, transformed, and enhanced before performing statistical analysis and visualization.

6.1 Data Pre-Processing Activities

Step	Description	Python Function(s) Used
Dataset Loading	Imported the CSV file into a Pandas DataFrame.	<code>pd.read_csv()</code>
Initial Data Inspection	Examined dataset structure, column names, and data types.	<code>head()</code> , <code>info()</code> , <code>shape</code>
Statistical Summary	Generated descriptive statistics for numerical columns.	<code>describe()</code>
Missing Value Analysis	Checked for null or missing values across all columns.	<code>isnull().sum()</code>
Duplicate Record Detection	Identified and removed duplicate transaction records.	<code>duplicated()</code> , <code>drop_duplicates()</code>
Data Type Validation	Verified and standardized appropriate data types.	<code>astype()</code> , <code>pd.to_datetime()</code>
Date & Time Conversion	Converted date columns into datetime format for analysis.	<code>pd.to_datetime()</code>
Feature Engineering	Created new analytical columns such as Month, Year, Sales Category, Average Item Price, and Customer Type.	<code>dt.month</code> , <code>dt.year</code> , <code>np.where()</code> , <code>pd.cut()</code>
Data Validation	Confirmed the cleaned dataset was free from inconsistencies before analysis.	<code>info()</code> , <code>describe()</code>

6.2 Data Cleaning Workflow

Process	Purpose
Dataset Import	Load the revenue transaction dataset into Python.
Data Inspection	Understand dataset structure and identify potential issues.
Missing Value Check	Ensure data completeness before analysis.
Duplicate Removal	Eliminate redundant transaction records.
Data Type Standardization	Convert columns into appropriate data types.
Feature Engineering	Generate additional business-related features.
Final Validation	Verify the dataset is clean and analysis-ready.

6.3 Dataset Preparation & Analytical Structure

- Prepared the Revenue Transaction dataset for analysis by validating data quality, standardizing data types, and creating analytical features using Pandas.
- Enhanced the dataset with business-oriented attributes including **Month**, **Year**, **Sales Category**, **Average Item Price**, and **Customer Type** to improve reporting and trend analysis.

Feature Engineering

```
df['AverageItemPrice'] = df['TotalAmount'] / df['TotalItem']
```

```
df['CustomerType'] = np.where(df['UseLoyaltyCard']==True,  
                               'Loyal Customer'
```

```
                               'Regular Customer')
```

- Validated the processed dataset before performing exploratory data analysis, statistical analysis, and visualization.

Processed Dataset Metrics

- **Total Rows: 53,820**
- **Total Columns: 16**

6.4 Analytical Dataset Architecture

In this project, the Revenue Transaction dataset was transformed into a clean, analysis-ready Data Frame using Python and Pandas. The preprocessing workflow ensured that the dataset was accurate, consistent, and enriched with additional business metrics to support descriptive, statistical, and visual analysis.

Conceptual Classification of Attributes

Transactional Metrics (Quantitative Measures)

- Net Sales
- Tax
- Total Amount
- Total Item
- Average Item Price

Descriptive Entities (Categorical Dimensions)

- Outlet
- Items
- Payment Method
- Sales Category
- Customer Type

Temporal Attributes

- Date
- Time
- Month
- Year

6.2 Derived Analytical Measures & KPI Calculations

6.2.1 Sales Category Classification

Categorized each transaction into **Low**, **Medium**, and **High** sales segments based on the Total Amount. This classification helps identify revenue distribution patterns and compare transaction performance across different sales levels.

```
df['SalesCategory'] = pd.cut(  
    df['TotalAmount'],  
    bins=[0, 100000, 300000, 1000000],  
    labels=['Low', 'Medium', 'High'])
```

6.2.2 Average Item Price Calculation

Calculated the average selling price per item by dividing the total transaction amount by the total number of items purchased. This KPI provides insights into customer purchasing behavior and product pricing trends.

```
df['AverageItemPrice'] = (  
    df['TotalAmount'] /  
    df['TotalItem'])
```

6.2.3 Customer Type Classification

Derived customer categories based on loyalty card usage to distinguish between **Loyal Customers** and **Regular Customers**. This KPI supports customer segmentation and helps evaluate the impact of loyalty programs on sales.

```
df['CustomerType'] = np.where(  
    df['UseLoyaltyCard'] == True,  
    'Loyal Customer',  
    'Regular Customer')
```

6.2.4 Temporal Feature Extraction

Extracted the **Month** and **Year** from the transaction date to enable monthly and yearly sales trend analysis, seasonal comparisons, and time-based reporting.

```
df['Month'] = df['Date'].dt.month  
df['Year'] = df['Date'].dt.year
```

7. Descriptive Statistics & Statistical Dispersion

Central tendency and dispersion measures were computed on the primary numerical attributes of the Revenue Transaction dataset to understand the distribution, variability, and business significance of each metric.

Metric Column	Mean (μ)	Median	Mode	Std. Dev (σ)	Dispersion & Business Interpretation
Net Sales	22,016.60	20,720.72	27,027.03	10,172.06	Represents the average revenue generated before tax. The relatively high standard deviation indicates considerable variation in transaction values across the dataset.
Tax	2,421.83	2,279.28	2,972.97	1,118.93	Shows the average tax collected per transaction. The moderate standard deviation suggests tax values vary in proportion to transaction amounts.
Total Amount	24,452.62	23,000.00	30,000.00	11,283.23	Represents the final customer payment. The high standard deviation indicates a wide distribution of transaction values and varying customer spending behaviour.
Total Item	1.15	1.00	1.00	0.36	Indicates that most transactions consist of a single item. The low standard deviation reflects consistent purchasing patterns across customers.
Average Item Price	21,278.75*	20,000.00*	30,000.00*	9,818.80*	Measures the average selling price per item. The observed variation reflects differences in product prices and transaction compositions.

Key Statistical Observations

- **Mean** represents the average value of each numerical attribute and provides an overview of business performance.
- **Median** identifies the central value and is less affected by outliers than the mean.
- **Mode** highlights the most frequently occurring value within each metric.
- **Standard Deviation** measures the spread of the data, indicating whether transaction values are consistent or highly variable.
- These statistical measures help assess sales trends, customer purchasing behaviour, pricing consistency, and revenue distribution across the dataset.

8. Exploratory Data Analysis (EDA) & Visualizations

Exploratory Data Analysis (EDA) was performed using **Python libraries (Matplotlib, Seaborn, and Plotly)** in **Google Colab** to identify revenue trends, customer purchasing patterns, payment method preferences, sales distribution, and key business insights. Various statistical and graphical techniques were applied to understand the dataset before drawing business conclusions.

8.1 Central Tendency & Distribution Analysis

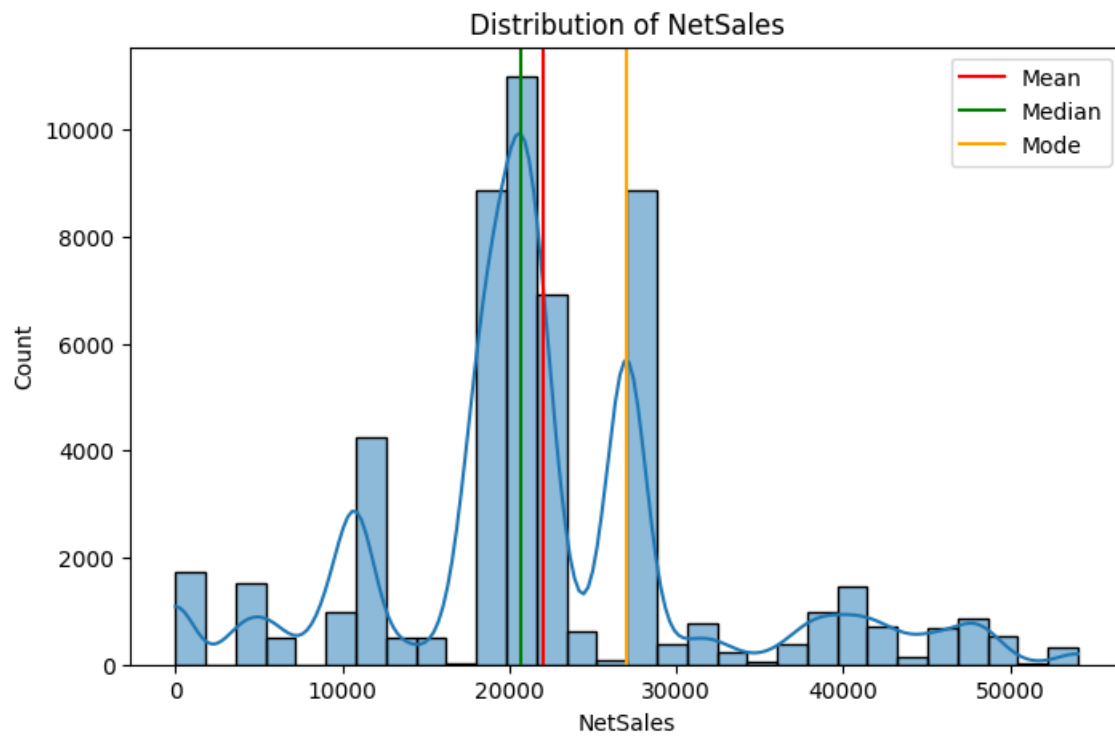
The distribution of key numerical variables was analyzed using histograms and central tendency measures (**Mean, Median, and Mode**). These visualizations help evaluate data distribution, detect

skewness, identify outliers, and understand customer transaction behaviour.

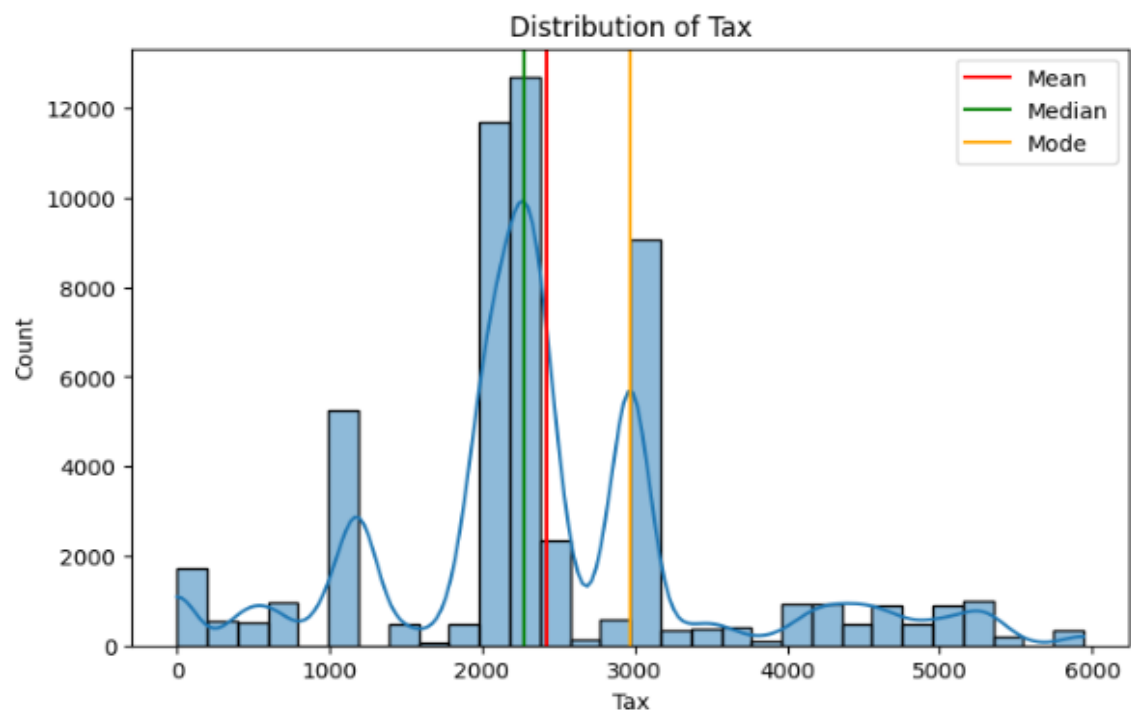
Business Interpretation

- The histogram illustrates the distribution of the selected numerical feature across all transactions.
- The **Mean** represents the average transaction value and provides an overall measure of central tendency.
- The **Median** indicates the middle value and is less affected by extreme values or outliers.
- The **Mode** identifies the most frequently occurring value within the dataset.
- The close proximity of the three measures suggests a relatively balanced distribution, while noticeable gaps may indicate skewness in transaction values.
- This analysis helps understand customer purchasing behavior and supports more accurate statistical interpretation of the revenue transaction dataset.

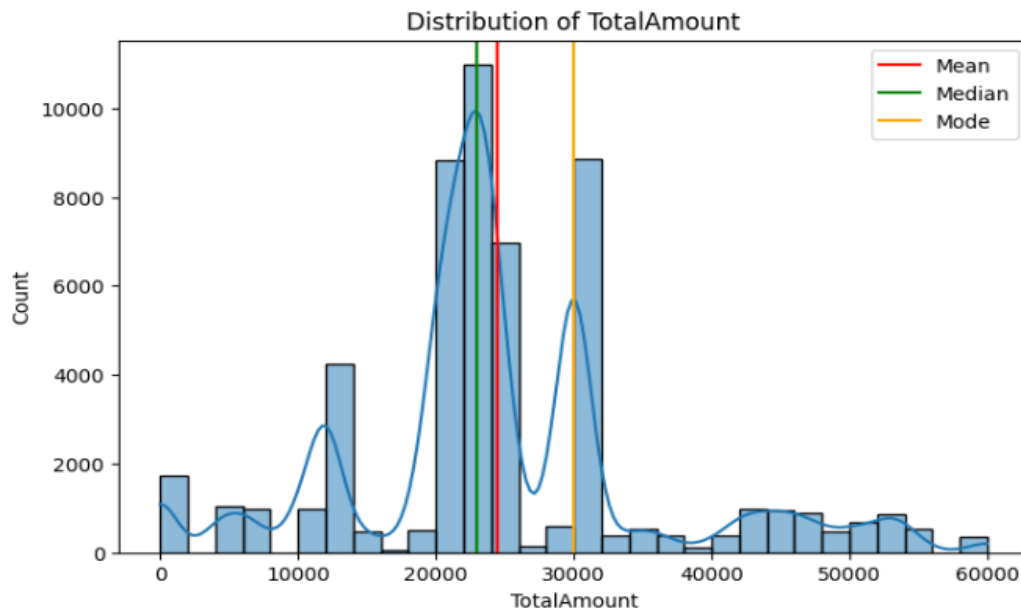
8.1.1 Net Sales Distribution



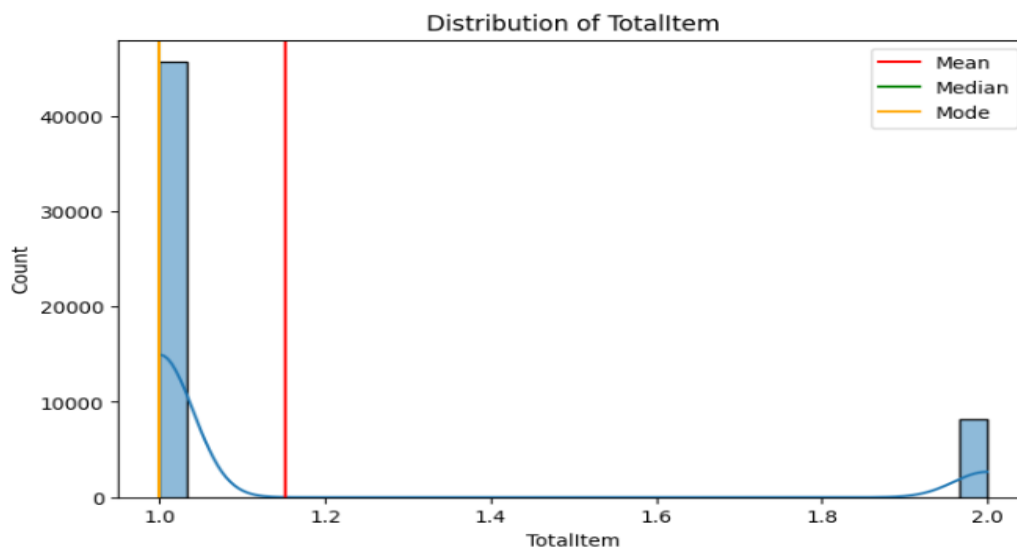
8.1.2 Tax Distribution



8.1.3 Total Amount Distribution



8.1.4 Total Item Distribution



Net Sales Distribution

- **Chart Type:** Histogram with KDE overlay.
- **Visual Elements:** Mean (Red), Median (Green), Mode (Orange).
- **Sales Pattern:** Most transactions are concentrated in the medium sales range.
- **Business Insight:** A few high-value transactions increase the overall average.

Key Finding: Revenue is mainly driven by medium-value sales.

Tax Distribution

- **Chart Type:** Histogram with KDE overlay.
- **Visual Elements:** Mean (Red), Median (Green), Mode (Orange).
- **Tax Pattern:** Tax values follow the sales distribution.
- **Business Insight:** Higher sales generate proportionally higher tax values.

Key Finding: Tax collection remains consistent with transaction values.

Total Amount Distribution

- **Chart Type:** Histogram with KDE overlay.
- **Visual Elements:** Mean (Red), Median (Green), Mode (Orange).
- **Spending Pattern:** Most customers make medium-value purchases.
- **Business Insight:** High-value purchases contribute significantly to total revenue.

Key Finding: Moderate purchases dominate overall sales.

Total Item Distribution

- **Chart Type:** Histogram with KDE overlay.
- **Visual Elements:** Mean (Red), Median (Green), Mode (Orange).
- **Purchase Pattern:** Most transactions contain one item.
- **Business Insight:** Bundle offers can help increase basket size.

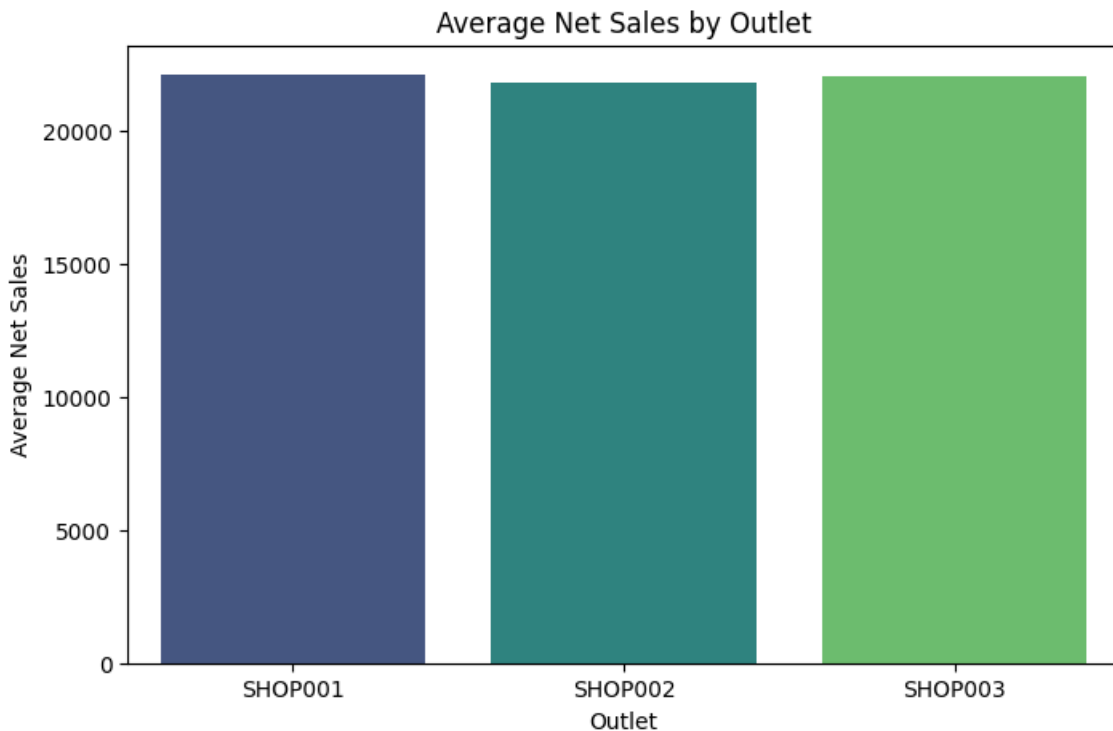
Key Finding: Single-item purchases are the most common.

Average Item Price Distribution

- **Chart Type:** Histogram with KDE overlay.
- **Visual Elements:** Mean (Red), Median (Green), Mode (Orange).
- **Pricing Pattern:** Most products fall within a consistent price range.
- **Business Insight:** Premium-priced items contribute additional revenue.

Key Finding: Product pricing is stable with occasional high-value items.

8.2 Average Net Sales by Outlet



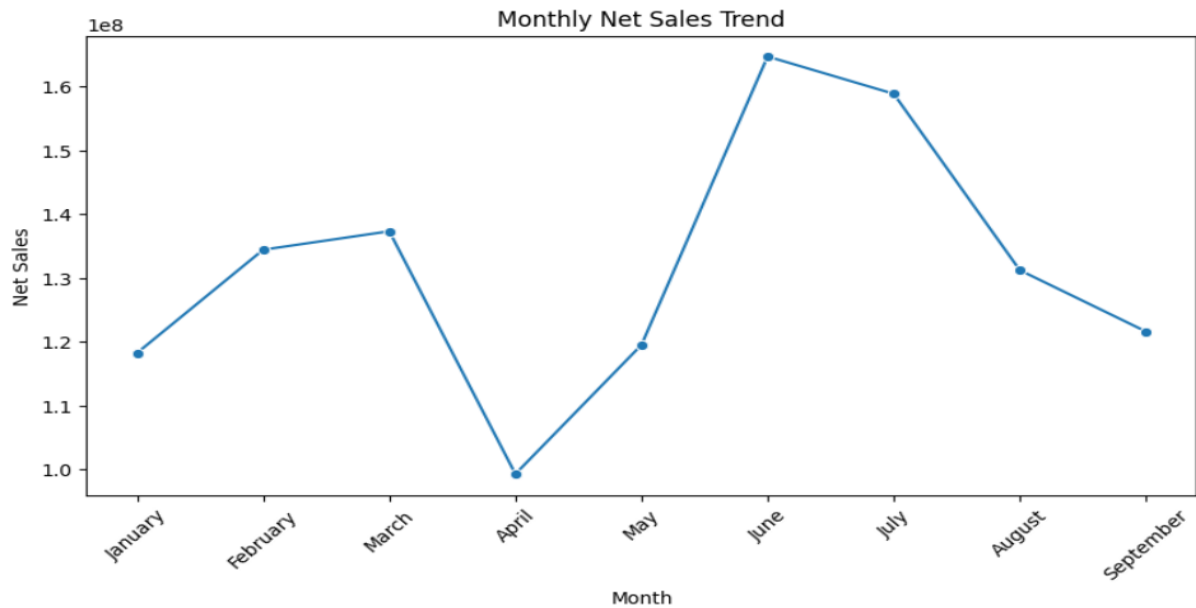
Interpretation

The bar chart compares the average net sales across the three retail outlets (SHOP001, SHOP002, and SHOP003). The analysis shows that the average net sales are very similar across all outlets, indicating consistent sales performance throughout the stores. Although SHOP001 records a slightly higher average net sales value, the difference compared to the other outlets is minimal. This suggests that revenue generation is well balanced across the retail network, with no significant performance gap between outlets. The findings indicate effective sales management and a uniform distribution of customer purchases across all store locations.

Business Insight

- All three outlets contribute almost equally to the overall revenue.
- SHOP001 has a marginally higher average net sales compared to the other outlets.
- No outlet is significantly underperforming, indicating balanced business operations.
- Similar sales performance across stores suggests consistent customer demand and standardized pricing strategies.
- Management can focus on increasing overall sales rather than prioritizing improvements for a specific outlet.

8.3 Monthly Net Sales Trend



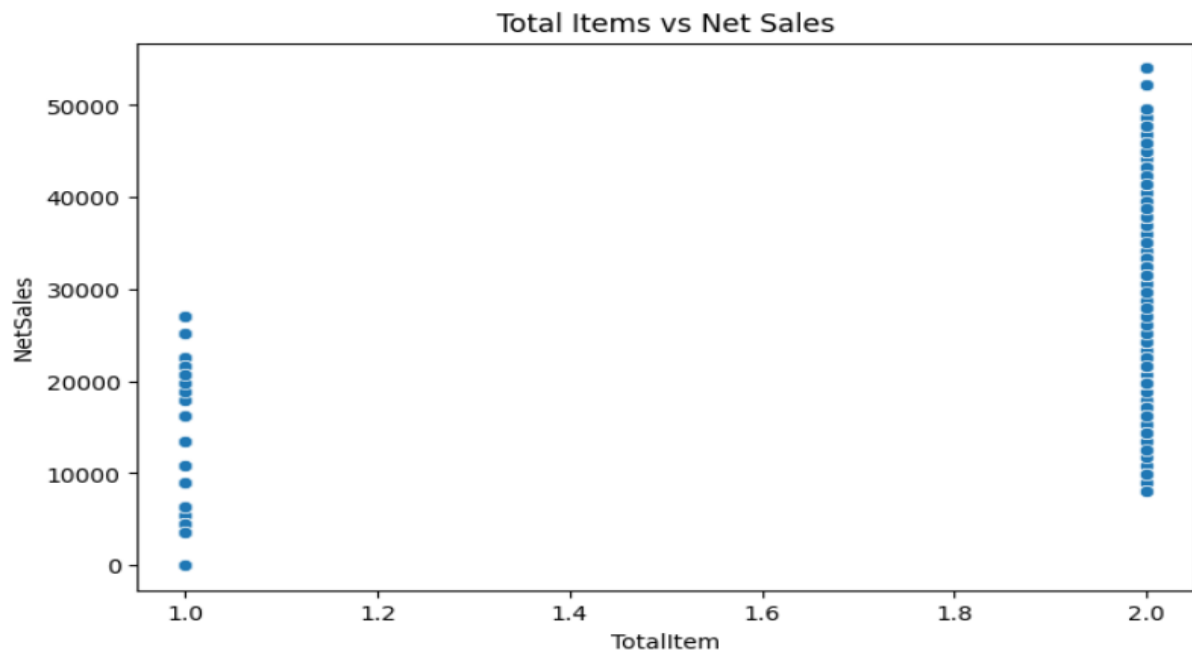
Interpretation

The line chart illustrates the monthly trend of net sales from January to September. Net sales increased steadily from January to March, followed by a significant decline in April, indicating a temporary reduction in sales performance. Sales recovered in May and reached the highest level in June, demonstrating a period of strong revenue generation. Although net sales declined slightly in July, they remained relatively high before decreasing further during August and September. Overall, the trend indicates seasonal fluctuations in sales with a clear peak during the middle of the year.

Business Insights

- Net sales increased consistently from January to March.
- April recorded the lowest monthly net sales, suggesting a temporary slowdown in business activity.
- June achieved the highest net sales, representing the best-performing month in the analysis period.
- Sales remained strong in July before gradually declining during August and September.
- Understanding these monthly patterns can help the business optimize inventory planning, promotional campaigns, and resource allocation during peak and low-demand periods.

8.4 Relationship Between Total Items and Net Sales



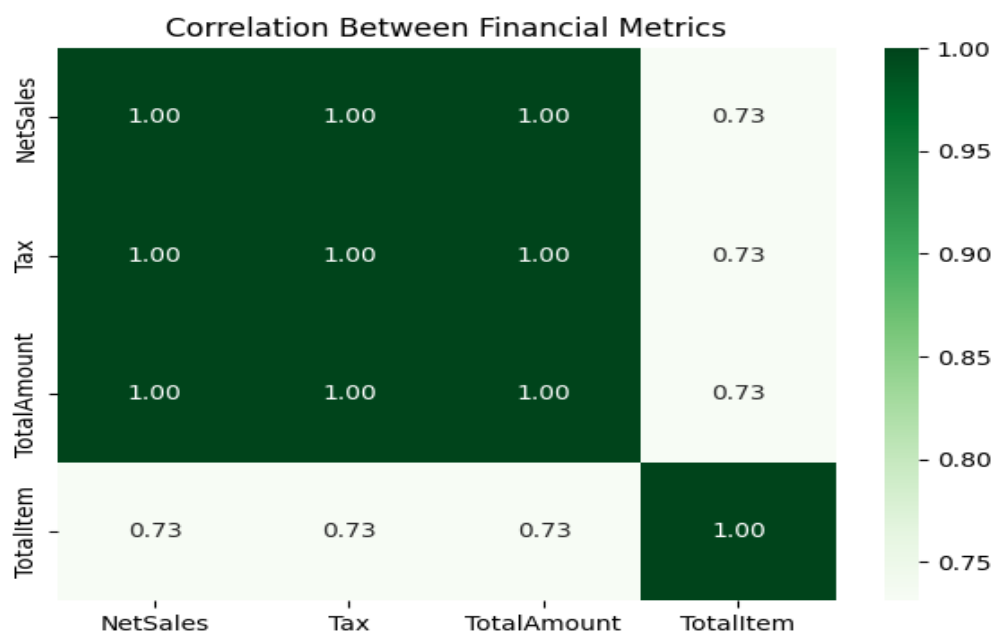
Interpretation

The scatter plot illustrates the relationship between the Total Items purchased and Net Sales for each transaction. Most transactions involve either one or two items, with transactions containing two items generally generating higher net sales than single-item purchases. However, there is noticeable variation in net sales within both groups, indicating that the transaction value depends not only on the number of items purchased but also on the price of the products. This suggests that higher-priced products contribute significantly to revenue, regardless of the quantity purchased.

Business Insights

- The majority of transactions consist of 1 or 2 items.
- Transactions with 2 items generally produce higher net sales than those with a single item.
- Net sales vary considerably within each item category, indicating differences in product prices.
- The number of items purchased alone is not a strong predictor of revenue; product value also plays an important role.
- Businesses can increase revenue by promoting premium products or encouraging customers to purchase additional high-value items.

8.5 Correlation Between Financial Metrics (Heatmap)



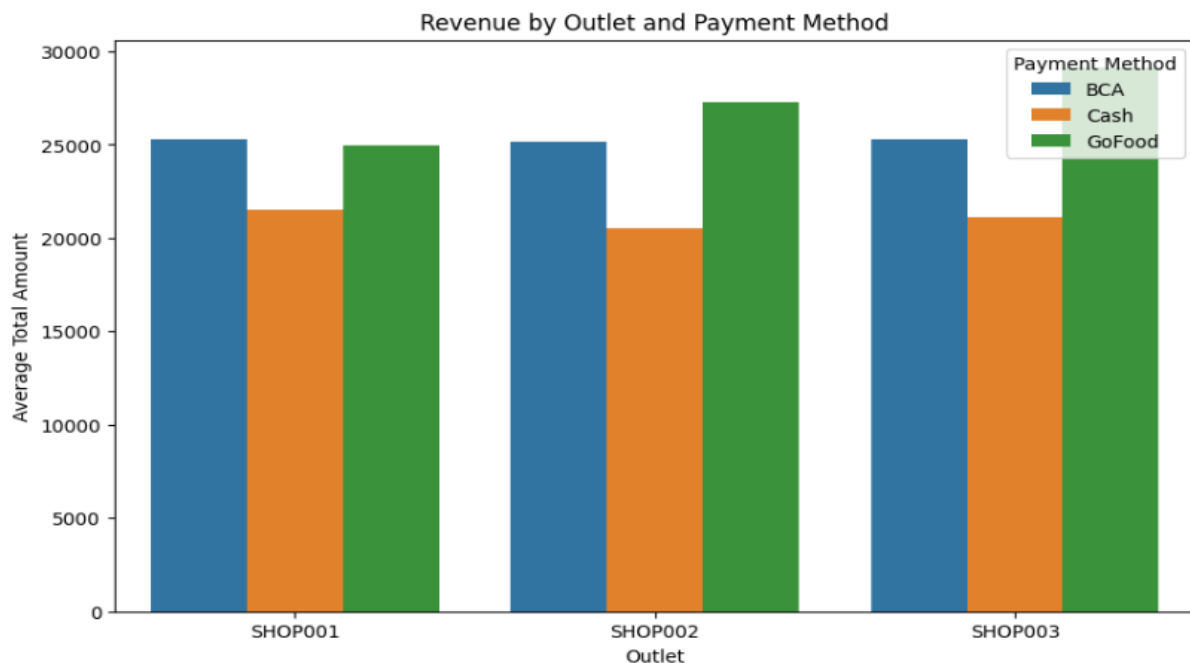
Interpretation

The correlation heatmap illustrates the strength of relationships between the key financial metrics: Net Sales, Tax, Total Amount, and Total Item. The results indicate a perfect positive correlation (1.00) among Net Sales, Tax, and Total Amount, showing that these variables increase together consistently. Total Item has a strong positive correlation (0.73) with the other financial metrics, suggesting that transactions involving more items generally result in higher sales and total transaction values. Overall, the heatmap confirms that the financial variables are closely related and contribute collectively to revenue generation.

Business Insights

- Net Sales, Tax, and Total Amount exhibit a perfect positive correlation (1.00), indicating that these metrics move together consistently.
- Total Item shows a strong positive correlation (0.73) with the financial metrics, meaning that purchasing more items generally leads to higher revenue.
- The strong relationships suggest that increasing transaction size can positively influence overall sales performance.
- The heatmap validates the consistency of the dataset and highlights key variables that drive business revenue.
- These insights can support pricing strategies, sales forecasting, and inventory planning.

8.6 Revenue Analysis by Outlet and Payment Method



Interpretation

The grouped bar chart compares the average total transaction amount across different outlets and payment methods. The results indicate that GoFood transactions generate the highest average revenue across all three outlets, while Cash transactions consistently record the lowest average transaction amount. BCA payments show relatively stable performance across all outlets, with average revenue falling between Cash and GoFood. Among the three outlets, SHOP002 records the highest average revenue for GoFood transactions, suggesting stronger customer engagement through digital food delivery services.

Business Insights

- GoFood is the highest-performing payment channel across all outlets, generating the greatest average transaction value.
- Cash transactions contribute the lowest average revenue, indicating lower-value purchases compared to digital payment methods.
- BCA payments maintain consistent average revenue across all outlets, reflecting stable customer usage.
- SHOP002 achieves the highest average revenue through GoFood, making it the strongest outlet for online orders.
- The business can increase revenue by promoting digital payment options, especially GoFood, and implementing targeted marketing campaigns to encourage higher-value transactions.

9. Strategic Business Recommendations

- **Increase High-Value Sales:** Promote premium products and bundle offers to improve the average transaction value.
- **Enhance Digital Payments:** Encourage customers to use digital payment methods through discounts and reward programs.
- **Seasonal Sales Planning:** Plan inventory and marketing campaigns based on monthly sales trends to maximize revenue.
- **Strengthen Customer Loyalty:** Improve loyalty programs to increase repeat purchases and customer retention.
- **Increase Basket Size:** Introduce cross-selling and combo offers to encourage customers to buy more items per transaction.
- **Monitor Outlet Performance:** Regularly track outlet-wise sales performance and adopt best practices across all stores.

10. Conclusion

This project successfully demonstrates an end-to-end Revenue Transaction Analysis using Python and data analytics techniques. The dataset was cleaned, transformed, and analysed using Exploratory Data Analysis (EDA), feature engineering, statistical analysis, and data visualization to uncover meaningful business insights.

The analysis identified revenue trends, customer purchasing patterns, outlet-wise sales performance, payment method preferences, and monthly sales variations. These insights can help businesses make informed decisions regarding sales strategies, customer engagement, inventory planning, and promotional activities.

Overall, this project highlights the importance of data-driven decision-making and demonstrates how Python-based analytics can support improved business performance, operational efficiency, and strategic planning.